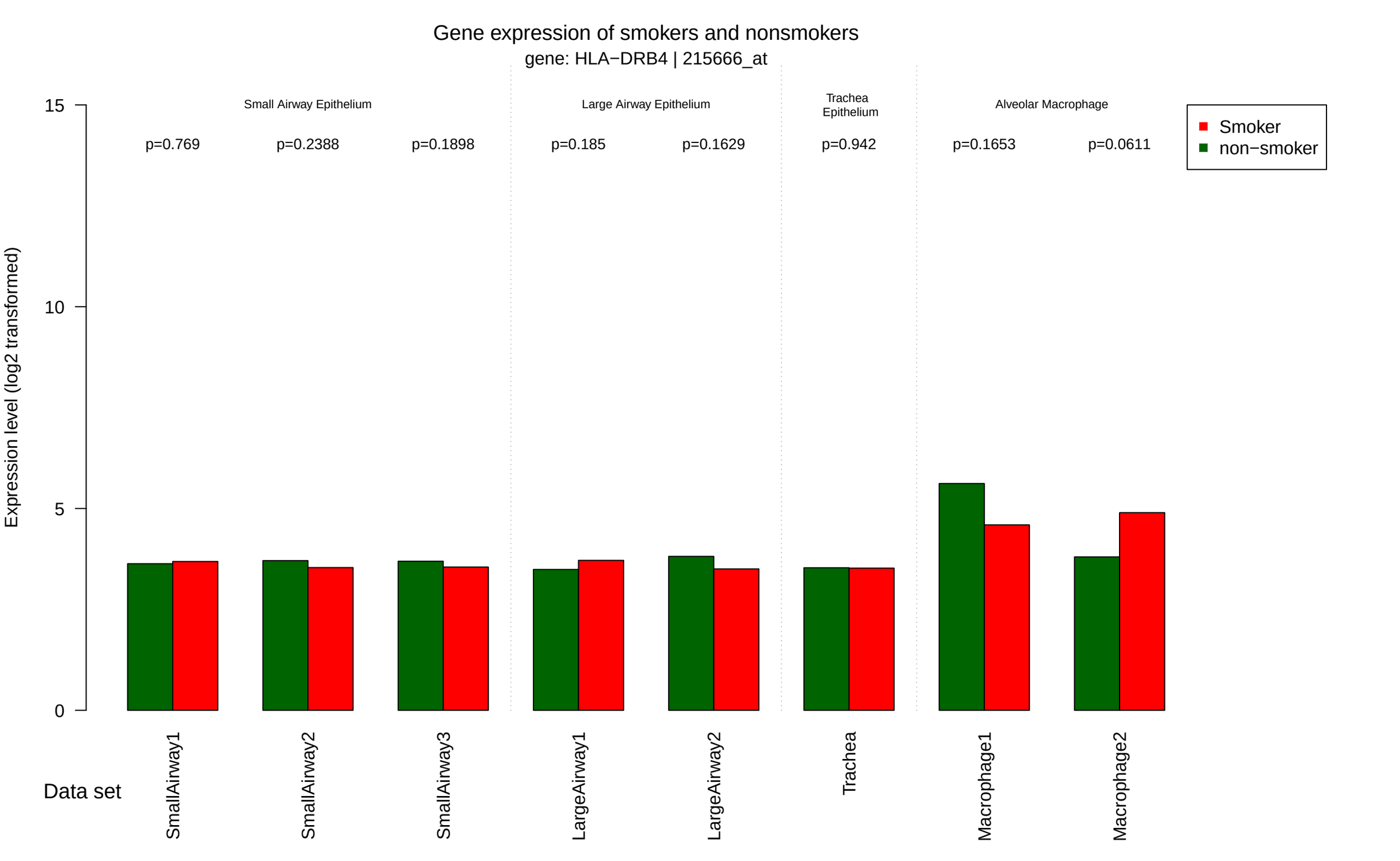
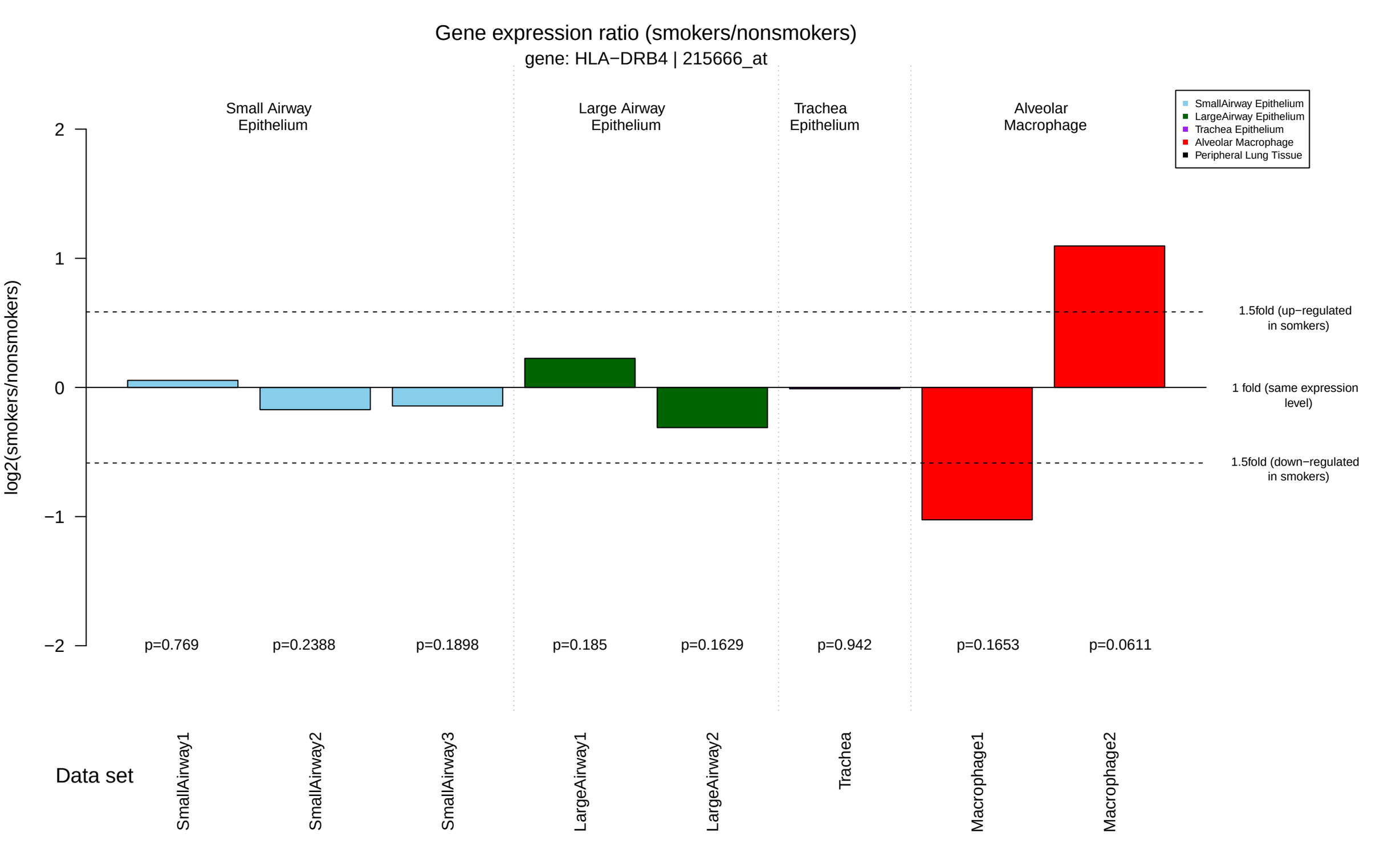
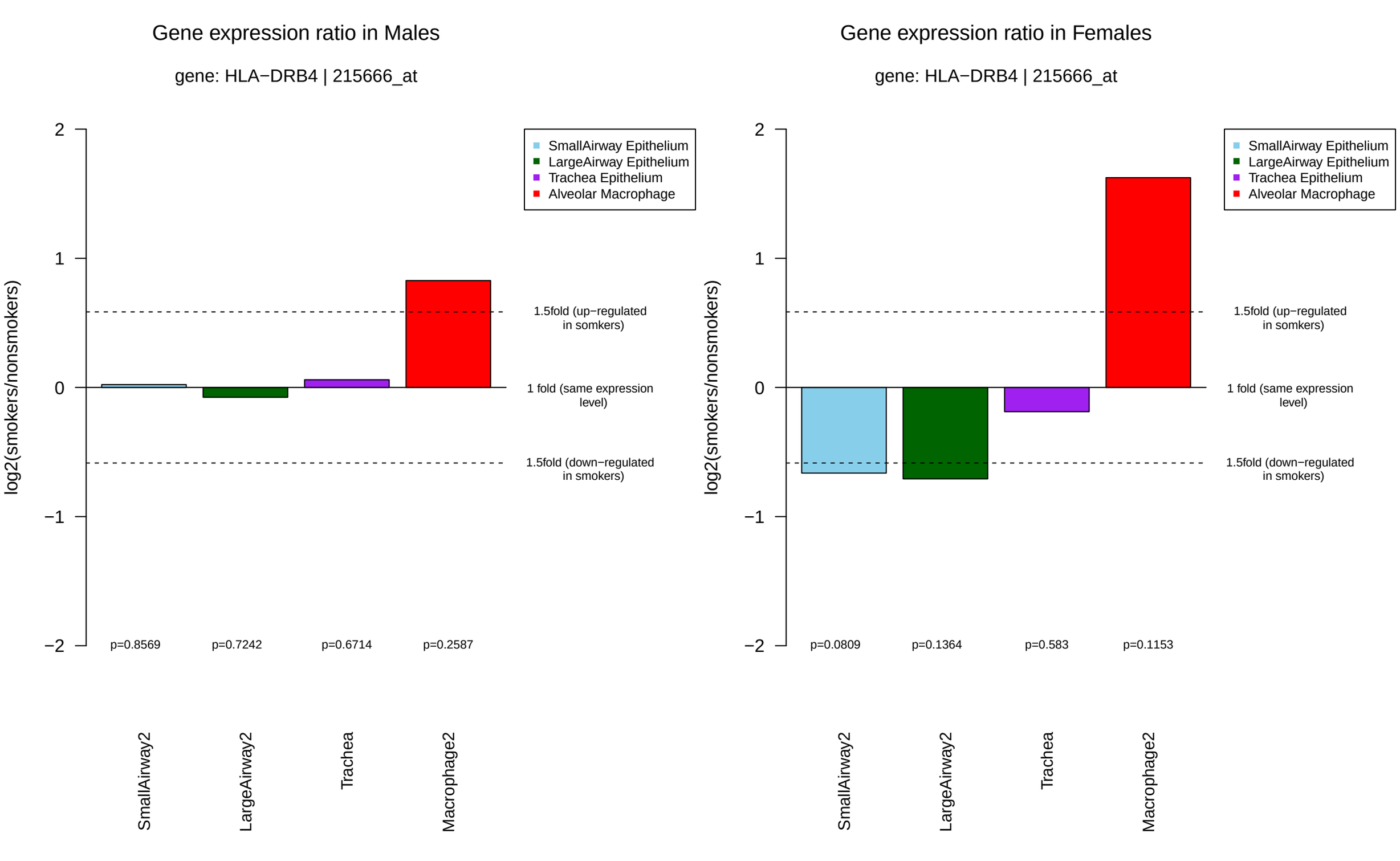
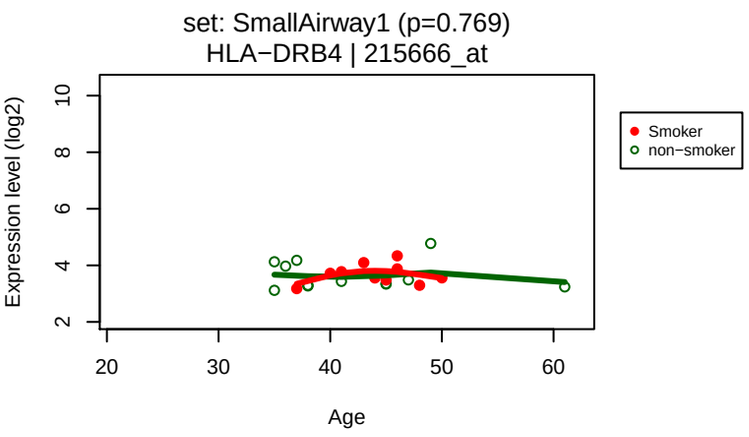
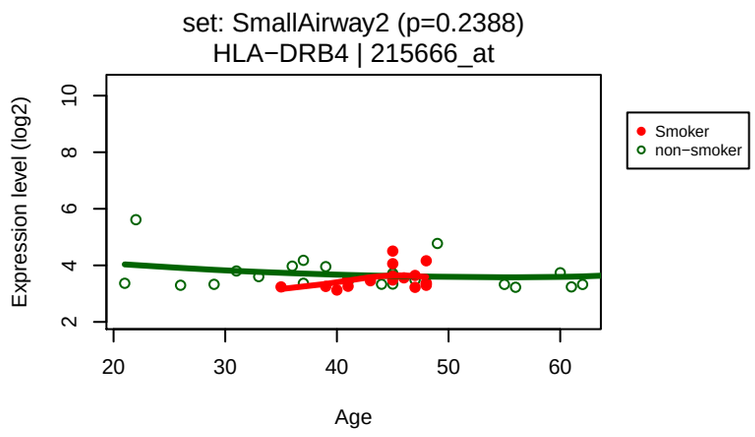
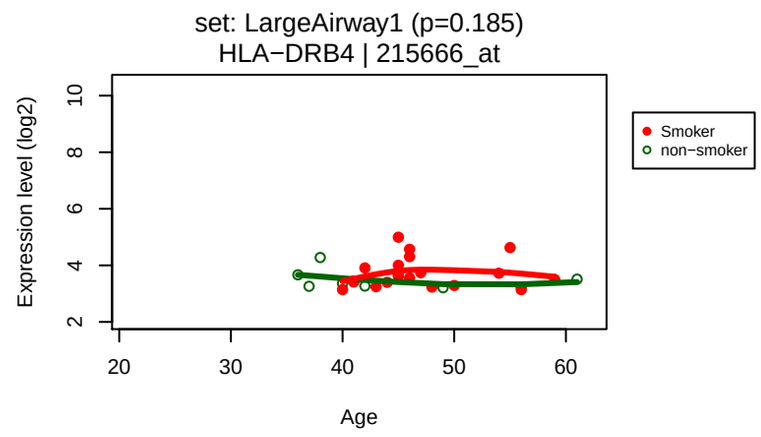

Gene expression of smokers and nonsmokers
gene: HLA−DRB4 | 215666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.769
p=0.2388
p=0.1898
p=0.185
p=0.1629
p=0.942
p=0.1653
p=0.0611
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

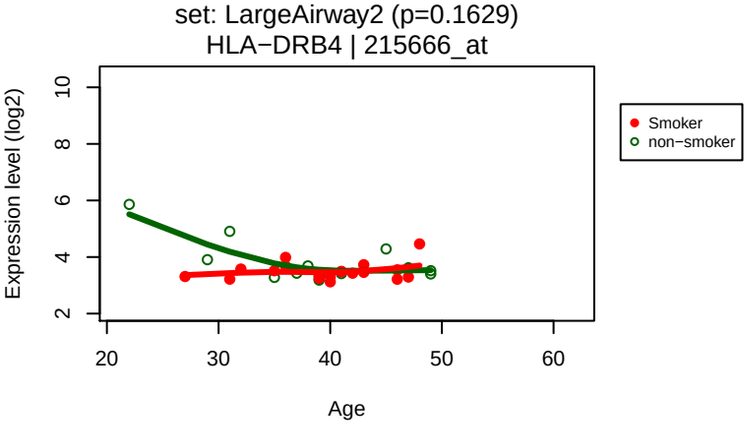

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB4 | 215666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.769
p=0.2388
p=0.1898
p=0.185
p=0.1629
p=0.942
p=0.1653
p=0.0611
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

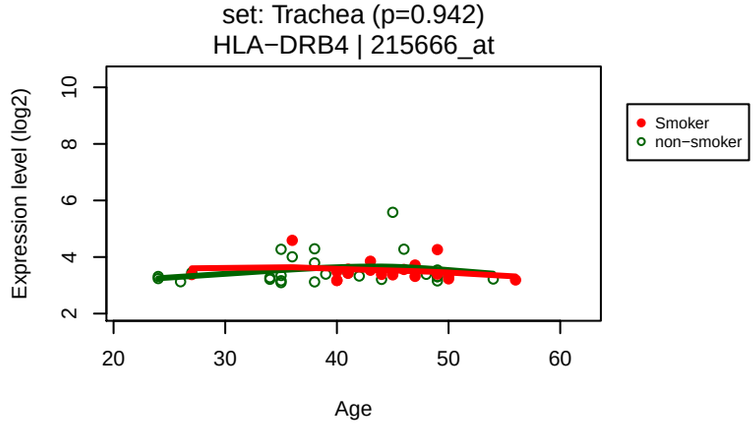

Gene expression ratio in Males
gene: HLA−DRB4 | 215666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8569
p=0.7242
p=0.6714
p=0.2587
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB4 | 215666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0809
p=0.1364
p=0.583
p=0.1153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

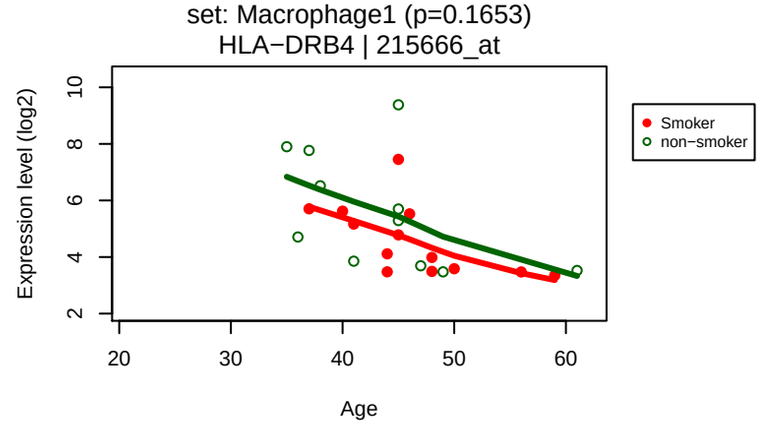

set: SmallAirway1 (p=0.769)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker

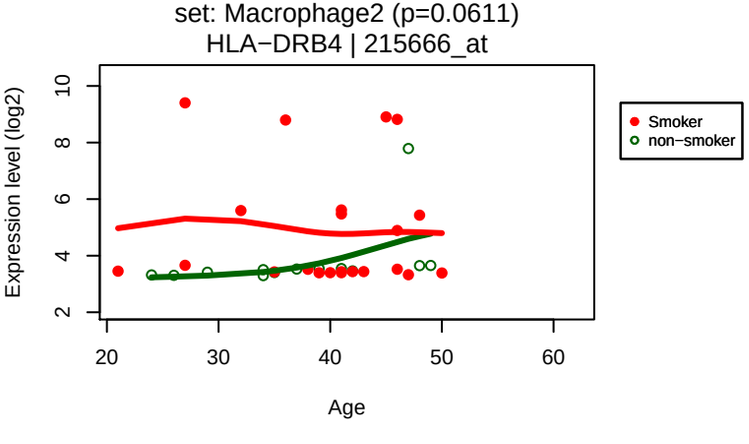

set: SmallAirway2 (p=0.2388)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker

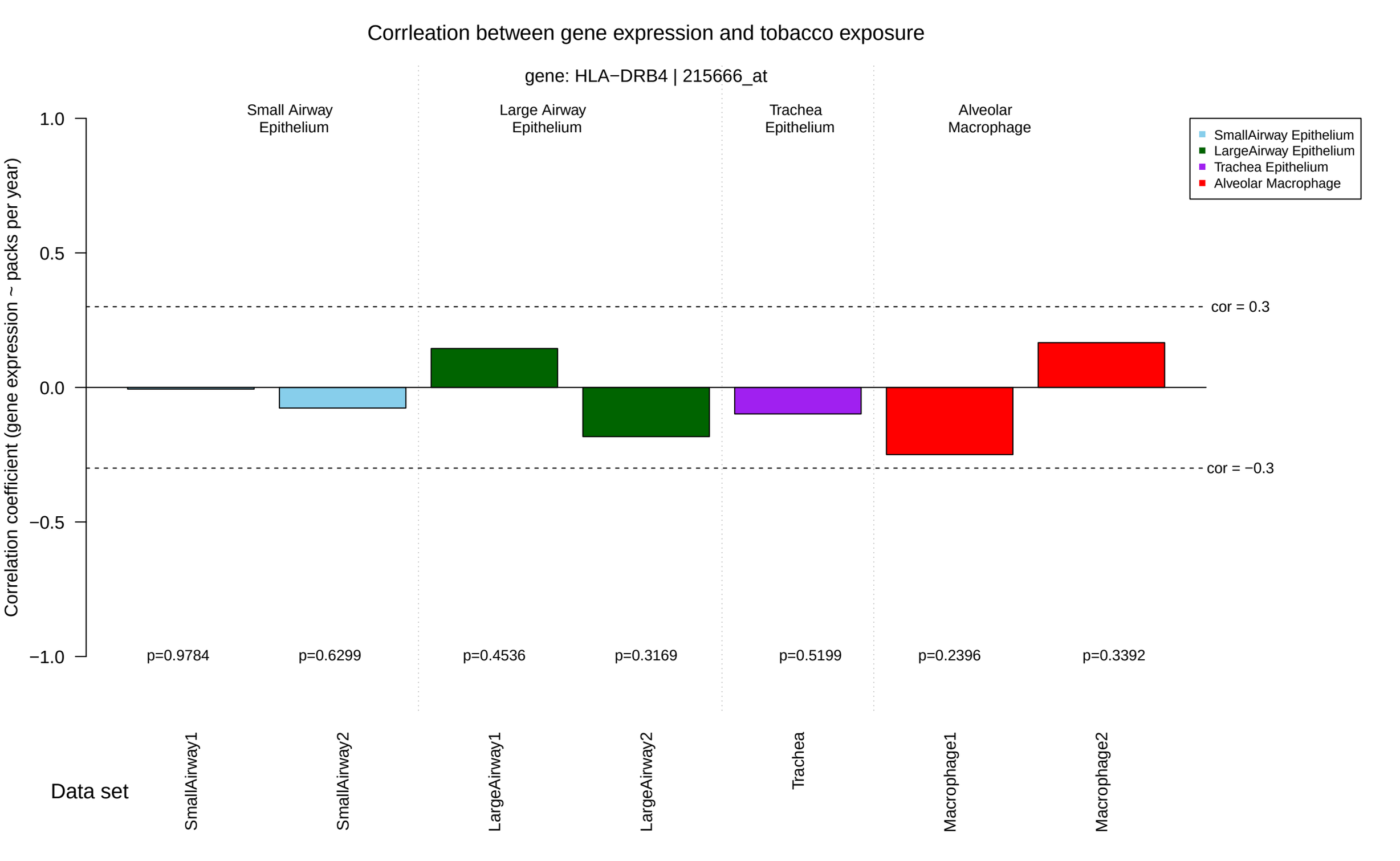

set: LargeAirway1 (p=0.185)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1629)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.942)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1653)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0611)
HLA−DRB4 | 215666_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB4 | 215666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9784
p=0.6299
p=0.4536
p=0.3169
p=0.5199
p=0.2396
p=0.3392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2